

On the joint distribution of runs in a sequence of multi-state trials

Masayuki Doi^a, Eiji Yamamoto^{b,*}

^a Department of Applied Mathematics, Okayama University of Science Graduated School, 1-1 Ridai-cho, Okayama 700, Japan

^b Department of Mathematical Information Science, Okayama University of Science, 1-1 Ridai-cho, Okayama 700, Japan

Received June 1997; received in revised form October 1997

Abstract

The joint distribution of c kinds of success runs in a sequence of $(c+1)$ -state trials is given by the finite Markov chain method proposed by Fu and Koutras (1994) and Fu (1996). © 1998 Elsevier Science B.V. All rights reserved

Keywords: Joint distribution; Markov chain; Multiple states; Run

1. Introduction

Studies of the run were started by Wald and Wolfowitz in the early 1940s. For more than 50 years after, the concept of runs has been extended by many researchers and used in many fields. In this article, we consider the joint distribution of multiple kinds of success runs.

The joint distribution of success and failure runs in a trinary sequence whose components are organised with a success, a failure or a tie was discussed by Doi et al. (1996). They started this study in conjunction with a cytologist Tsutsui who hoped to check the information on a particular part (scaffold attachment region, SAR) in a deoxyribonucleic acid (DNA) by evaluating the pattern of two bases A and T on SAR (Tsutsui et al., 1993). In this problem, they regarded a run (patch) of the specific bases of size greater than or equal to k on DNA sequence of the length n in SAR as success and failure runs. They found out the method of checking the information of bases in comparison with the joint distribution of the two-dimensional random vector $(G_{n,k_1}^{(1)}, G_{n,k_2}^{(2)})$, which counts the number of success and failure runs.

Here, this method will be modified for the problem to find out the joint distribution of a success run vector in sequence consisting of multiple kinds of successes and failures. This means that when c kinds of success $S_1, S_2, \dots, S_c$ and one kind of failure F exist, we get to give the joint distribution of a c -dimensional success run vector in $(c+1)$ -state trials.

To solve the problem, we employ the method proposed by Fu and Koutras (1994) and Fu (1996). In this method the objective statistic $X_{n,k}$ is imbedded into the finite Markov chain $\{Y_t : t \in \Gamma_n\}$ where $\Gamma_n =$

* Corresponding author.

$\{0, 1, \dots, n\}$, and the probability distribution $\Pr\{X_{n,k} = x\}$ is given by utilizing the state distribution of Y_t . Here, the statistic $X_{n,k}$ means the number of success runs, which is classified into four kinds as follows:

- $N_{n,k}$: the number of non-overlapping consecutive k successes, in the sense of Feller's (1968) counting,
- $G_{n,k}$: the number of success runs of size greater than or equal to k ,
- $M_{n,k}$: the number of overlapping consecutive k successes, in the sense of Ling's (1988) counting,
- $E_{n,k}$: the number of success runs of size exactly k , in the sense of Mood's (1940) counting.

Random variable vectors consisting of these four statistics can be generally imbedded into the finite Markov chain, and for convenience we note four random variable vectors as follows,

$$N = (N_{n,k_1}^{(1)}, N_{n,k_2}^{(2)}, \dots, N_{n,k_c}^{(c)}),$$

$$G = (G_{n,k_1}^{(1)}, G_{n,k_2}^{(2)}, \dots, G_{n,k_c}^{(c)}),$$

$$M = (M_{n,k_1}^{(1)}, M_{n,k_2}^{(2)}, \dots, M_{n,k_c}^{(c)}),$$

$$E = (E_{n,k_1}^{(1)}, E_{n,k_2}^{(2)}, \dots, E_{n,k_c}^{(c)}).$$

The following section discusses the joint distribution of multiple kinds of success runs as mentioned above.

2. The joint distribution of multiple kinds of runs

2.1. Definitions for joint distributions

Doi et al. (1996) showed joint distributions of success and failure runs in a trinary sequence for the two statistics $N_{n,k}$ and $G_{n,k}$. Here, we extend them generally in c kinds of success runs in a $(c+1)$ -ary sequence for four vectors N, G, M and E .

For given n , let $\Gamma_n = \{0, 1, \dots, n\}$ be an index set, $\Omega = \{a_{(r_1, r_2, \dots, r_c)} : r_j = 1, 2, \dots, m_j; j = 1, 2, \dots, c\}$ be a finite state space and $\Xi = \{(x_1, x_2, \dots, x_c) : x_j = 0, 1, \dots, l_j; j = 1, 2, \dots, c\}$ be a c -dimensional sample space. For convenience, we use a random vector $X = (X_{n,k_1}^{(1)}, X_{n,k_2}^{(2)}, \dots, X_{n,k_c}^{(c)})$ to represent any of the four random vectors N, G, M , and E .

Definition 1. A nonnegative integer random vector X can be imbedded into a finite Markov chain if

1. there exists a finite Markov chain $\{Y_t : t \in \Gamma_n\}$ defined on the finite state space Ω ,
2. there exists a finite partition $\{C_x : x \in \Xi\}$ on the state space Ω ,
3. for every $x \in \Xi$, we have $\Pr\{X = x\} = \Pr\{Y_n \in C_x\}$.

Let $Z(1), Z(2), \dots, Z(n)$ be a sequence of $(c+1)$ -state trials such that $Z(t) = j$ ($j = 1, \dots, c$) or 0 if the t th trial results the j th kind of success S_j or the failure F , respectively. Further, we define $x = (x_1, x_2, \dots, x_c)$ a vector with the number of the j th kind of success runs, x_j , and (j, i_j) a realization of an ending block random vector $E_t = (Z(t), i_{Z(t)})$, where i_j is the number of consecutive successes of the j th kind before being accepted as a run including states $i_j = -1$ and/or $i_j = -2$ for the statistics G, M and E , and $i_0 \equiv 0$ for the outcome F ($j = 0$).

Then we may correspond the element of the state space Ω , $a_{(r_1, r_2, \dots, r_c)}$ to a 2-tuple $(x, (j, i_j))$, i.e. $a_{(r_1, r_2, \dots, r_c)} = (x, (j, i_j))$. Therefore, Ω can be formed as $\Omega = \{(x, (j, i_j)) : x \in \Xi; (j, i_j) \in J\}$, where J is a set of realization of E_t . The event $Y_t = (X, E_t)$ means that, in a sequence of outcomes up to the t th trial, x_j , $j = 1, 2, \dots, c$ runs have occurred and the latter i_j outcomes are S_j 's following other kind success $S_{j'}$ ($j' \neq j$) or a failure F at the $(t - i_j)$ th outcome.

For any $(x, (j, i_j)) \in \Omega$, we define $\pi_t(x, j, i_j) = \Pr\{Y_t = (x, (j, i_j))\}$, then the recursive relationship $\{\pi_{t+1}(x, j', i_{j'})\} = g(\{\pi_t(x, j, i_j)\})$ can be established. Let $\mathbf{0}$ be a null vector and we set the probability at the initial time, $\pi_0(\mathbf{0}, 0, 0) = 1$. For finding out the distribution of the random vector X , we can use the

following equation such as:

$$\Pr\{X = \mathbf{x}\} = \sum_{(\mathbf{x}, (j, i_j)) \in C_{\mathbf{x}}} \pi_n(\mathbf{x}, j, i_j), \quad \mathbf{x} \in \Xi.$$

The transition process of probabilities has several cases and is noted as follows:

Developing a run. The case of S_j coming after $\dots \overbrace{S_j S_j \dots S_j}^{i_j (< k_j - 1)}$.

Achieving of developing a run. The case of S_j coming after $\dots \overbrace{S_j S_j \dots S_j}^{k_j - 1}$.

Terminating of developing a run. The case of $S_{j'}$ ($j \neq j'$) or F coming after $\dots \overbrace{S_j S_j \dots S_j}^{i_j (< k_j - 1)}$.

Waiting states (G, M). The case of S_j coming after $\dots \overbrace{S_j S_j \dots S_j}^{i_j (\geq k_j)}$.

Waiting states (E). The case of $\dots \overbrace{S_j S_j \dots S_j}^{k_j}$.

Overflow states (E). The case of $\dots \overbrace{S_j S_j \dots S_j}^{i_j (> k_j)}$.

2.2. The distribution of N

Let us consider the state space

$$\Omega = \{(\mathbf{x}, (j, i_j)) : \mathbf{x} \in \Xi, (j, i_j) \in J\},$$

where $\Xi = \{\mathbf{x} : \mathbf{x} = (0, 0, \dots, 0), \dots, ([n/k_1], [n/k_2], \dots, [n/k_c])\}$ and $J = \{(j, i_j) : i_0 = 0, i_j = 0, 1, 2, \dots, k_j -$

$1, j = 0, 1, 2, \dots, c\}$. For any sequence of outcomes of the length t and $j = 1, 2, \dots, c, \dots \overbrace{S_j S_j \dots S_j}^{m_j}$, let m_j be the number of trailing successive S_j 's counting backwards. We define $Y_t = (\mathbf{x}, (j, i_j))$ when there exist x_j nonoverlapping consecutive k_j successes and $i_j \equiv m_j \pmod{k_j}$. For given x_j ($0 \leq x_j \leq [n/k_j]$), we introduce the subset $C_{\mathbf{x}} = \{(\mathbf{x}, (j, i_j)) : (j, i_j) \in J\}$. Clearly, $\{C_{\mathbf{x}} : \mathbf{x} \in \Xi\}$ forms a partition of the state space Ω and the equation $\Pr\{N = \mathbf{x}\} = \Pr\{Y_n \in C_{\mathbf{x}}\}$ holds. On the assumption that events occur independently and the j th kind success appears with the probability $p_j(t)$ ($j = 1, 2, \dots, c$) or the failure does with $q(t)$ ($= 1 - (p_1(t) + p_2(t) + \dots + p_c(t))$) at time t , the recursive formula $\{\pi_{t+1}(\mathbf{x}, j', i_{j'})\} = g(\{\pi_t(\mathbf{x}, j, i_j)\})$ becomes as follows:

Developing a run

$$\pi_{t+1}(\mathbf{x}, j, i_j + 1) = \pi_t(\mathbf{x}, j, i_j) \cdot p_j(t + 1), \quad i_j = 1, 2, \dots, k_j - 2.$$

Achievement of developing a run

$$\pi_{t+1}(\mathbf{x} + \mathbf{e}_j, j, 0) = \pi_t(\mathbf{x}, j, k_j - 1) \cdot p_j(t + 1).$$

Termination of developing a run

$$\pi_{t+1}(\mathbf{x}, j', 1) = \sum_{j \neq j'} \sum_{i_j=0}^{k_j-1} \pi_t(\mathbf{x}, j, i_j) \cdot p_{j'}(t + 1),$$

$$\pi_{t+1}(\mathbf{x}, 0, 0) = \sum_{j=1}^c \sum_{i_j=0}^{k_j-1} \pi_t(\mathbf{x}, j, i_j) \cdot q(t + 1),$$

where e_j is a unit vector with 1 in the j th element and 0 otherwise. By using these formulae, the distribution of N can be solved with the state distribution of Y_n as

$$\Pr\{N = x\} = \sum_{(x, (j, i_j)) \in C_x} \pi_n(x, j, i_j).$$

2.3. The distribution of G

Let us consider the state space

$$\Omega = \{(x, (j, i_j)) : x \in \Xi, (j, i_j) \in J\} - \bigcup_{j=1}^c \{(\bar{x}_j, (j, -1))\},$$

where $\Xi = \{x : x = (0, 0, \dots, 0), \dots, [(n+1)/(k_1+1)], [(n+1)/(k_2+1)], \dots, [(n+1)/(k_c+1)]\}$ and $J = \{(j, i_j) : i_0 = 0, i_j = -1, 0, 1, 2, \dots, k_j - 1, j = 1, 2, \dots, c\}$, and $\bar{x}_j$ is a vector with 0 in the j th element such as $\bar{x}_j = x - x_j e_j = (x_1, x_2, \dots, x_{j-1}, 0, x_{j+1}, \dots, x_c)$. The finite Markov chain $\{Y_t : t \in \Gamma_n\}$ defined on Ω

is given as follows. For any sequence of outcomes of the length $t, \dots, \overbrace{S_j S_j \dots S_j}^{m_j}$, we define $Y_t = (x, (j, i_j))$ when there exist $x_j \geq 0, j = 1, 2, \dots, c$ success runs of the size k_j exactly before the last $m_j + 1$ outcomes

$\overbrace{S_j \dots S_j}^{m_j}$ (or F) $\overbrace{S_j S_j \dots S_j}^{m_j}$, and $0 \leq i_j = m_j \leq k_j - 1$. In addition, we define $Y_t = (x, (j, -1))$, which can be viewed as a waiting state, when $m_j \geq k_j$ and $x_j \geq 1$. There are $(x_j - 1)$ success runs of the j th kind before the last $(m_j + 1)$ outcomes. Note that the condition $m_j \geq k_j$ implies that x_j must be equal or greater than 1; that is, the state $(\bar{x}_j, (j, -1))$ does not make any sense.

For given $x_j (0 \leq x_j \leq [(n+1)/(k_j+1)], j = 1, 2, \dots, c)$, we introduce the subset $C_x = \{(x, (j, i_j)) : j = 0, 1, \dots, c\}$. Clearly, $\{C_x : x \in \Xi\}$ forms a partition of the state space Ω and $\Pr\{G = x\} = \Pr\{Y_n \in C_x\}, x \in \Xi$. Under the same condition as the previous subsection, the recursive formula $\{\pi_{t+1}(x, j', i_{j'})\} = g(\{\pi_t(x, j, i_j)\})$ becomes as follows:

Developing a run

$$\pi_{t+1}(x, j, i_j + 1) = \pi_t(x, j, i_j) \cdot p_j(t + 1), \quad i_j = 1, 2, \dots, k_j - 2.$$

Achievement of developing a run

$$\pi_{t+1}(x + e_j, j, -1) = \pi_t(x, j, k_j - 1) \cdot p_j(t + 1).$$

Waiting states

$$\pi_{t+1}(x, j, -1) = \pi_t(x, j, -1) \cdot p_j(t + 1).$$

Termination of developing a run

$$\begin{aligned} \pi_{t+1}(x, j', 1) &= \sum_{j \neq j'} \sum_{i_j=-1}^{k_j-1} \pi_t(x, j, i_j) \cdot p_{j'}(t + 1). \\ \pi_{t+1}(x, 0, 0) &= \sum_{j=1}^c \sum_{i_j=-1}^{k_j-1} \pi_t(x, j, i_j) \cdot q(t + 1). \end{aligned}$$

By paying attention to these transition processes, the distribution of $\mathbf{G}$ can be given by the state distribution of Y_n as

$$\Pr\{\mathbf{G} = \mathbf{x}\} = \sum_{(\mathbf{x}, (j, i_j)) \in C_x} \pi_n(\mathbf{x}, j, i_j).$$

2.4. The distribution of M

Let us consider the state space

$$\Omega = \{(\mathbf{x}, (j, i_j)) : \mathbf{x} \in \Xi\} - \bigcup_{j=1}^c \bigcup_{x_j=0}^{n-k_j} \{(\mathbf{x}, j, -1)\},$$

where $\Xi = \{\mathbf{x} : \mathbf{x} = (0, 0, \dots, 0), \dots, (n - k_1 + 1, n - k_2 + 1, \dots, n - k_c + 1)\}$ and $\mathbf{J} = \{(j, i_j) : i_0 = 0, i_j = -1, 0, 1, \dots, k_j - 1, j = 1, 2, \dots, c\}$. The finite Markov chain $\{Y_t : t \in \Gamma_n\}$ defined on Ω is given as follows. Let x_j be the number of overlapping success runs of the j th kind (Ling's counting) of the length k_j up to the t th trial. We define $Y_t = (\mathbf{x}, (j, i_j))$ when $i_j \leq k_j - 1$, and $Y_t = (\mathbf{x}, (j, -1))$ as a waiting state when there are more than k_j executive successes of the j th kind. The corresponding partition of the state space Ω is specified by

$$C_x = \begin{cases} \{(\mathbf{x}, (j, i_j)) : i_j = -1, 0, 1, \dots, k_j - 1\}, & x_j = 1, \dots, l_j - 1, \\ \{(\mathbf{x}, (j, -1))\}, & x_j = l_j, \end{cases}$$

where $l_j = n - k_j + 1$. Clearly, it holds that $\Pr\{\mathbf{M} = \mathbf{x}\} = \Pr\{Y_n \in C_x\}$. Subsequently, the recursive formula $\{\pi_{t+1}(\mathbf{x}, j', i_{j'})\} = g(\{\pi_t(\mathbf{x}, j, i_j)\})$ becomes as follows:

Developing a run

$$\pi_{t+1}(\mathbf{x}, j, i_j + 1) = \pi_t(\mathbf{x}, j, i_j) \cdot p_j(t + 1), \quad i_j = 1, 2, \dots, k_j - 2.$$

Achievement of developing a run

$$\pi_{t+1}(\mathbf{x} + \mathbf{e}_j, j, -1) = \pi_t(\mathbf{x}, j, k_j - 1) \cdot p_j(t + 1).$$

Waiting states

$$\pi_{t+1}(\mathbf{x} + \mathbf{e}_j, j, -1) = \pi_t(\mathbf{x}, j, -1) \cdot p_j(t + 1).$$

Termination of developing a run

$$\begin{aligned} \pi_{t+1}(\mathbf{x}, j', 1) &= \sum_{j \neq j'} \sum_{i_j=-1}^{k_j-1} \pi_t(\mathbf{x}, j, i_j) \cdot p_{j'}(t + 1), \\ \pi_{t+1}(\mathbf{x}, 0, 0) &= \sum_{j=1}^c \sum_{i_j=-1}^{k_j-1} \pi_t(\mathbf{x}, j, i_j) \cdot q(t + 1). \end{aligned}$$

By paying attention to these transition processes, the distribution of $\mathbf{M}$ can be given by the state distribution of Y_n as

$$\Pr\{\mathbf{M} = \mathbf{x}\} = \sum_{(\mathbf{x}, (j, i_j)) \in C_x} \pi_n(\mathbf{x}, j, i_j).$$

2.5. The distribution of E

Let us consider the state space

$$\Omega = \{(x, (j, i_j)) : x \in \Xi\} - \bigcup_{j=1}^c \{(\bar{x}_j, (j, -2))\},$$

where $\Xi = \{x : x = (0, 0, \dots, 0), \dots, [(n+1)(k_1+1)], [(n+1)(k_2+1)], \dots, [(n+1)(k_c+1)]\}$ and $J = \{(j, i_j) : i_0 = 0, i_j = -2, -1, 0, 1, 2, \dots, k_j - 1, j = 1, 2, \dots, c\}$. The finite Markov chain $\{Y_t : t \in \Gamma_n\}$ defined on Ω is given as follows. A success run of the j th kind whose length becomes greater than k_j does not count as a success run for E . The states $(x, (j, -1))$, $(x, (j, -2))$ are used to indicate such states. Specifically, in a

sequence of outcomes $\dots \overbrace{S_j \dots S_j}^{i_j}$, we define two following states.

- Overflow states: $(x, (j, -1))$, $x \in \Xi$. The event $Y_t = (x, (j, -1))$ means that there are more than $(k_j + 1)$ executive successes of the j th kind and exactly k_j success runs of the j th kind of size k_j appear before the last $(i_j + 1)$ outcomes.
- Waiting states: $(x, (j, -2))$, $x \in \Xi - \bigcup_{j=1}^c \{\bar{x}_j\}$. The event $Y_t = (x, (j, -2))$ means that exactly k_j executive successes of the j th kind have occurred.

It is easy to specify the partition of Ω , $C_x = \{(x, (j, i_j)) : i_j = -2, -1, 0, \dots, k_j - 1\}$, $x \in \Xi$. Following the notation in the previous sections to show the recursive formula $\{\pi_{t+1}(x, j', i_{j'})\} = g(\{\pi_t(x, j, i_j)\})$, then it will become as follows:

Developing a run

$$\pi_{t+1}(x, j, i_j + 1) = \pi_t(x, j, i_j) \cdot p_j(t + 1), \quad i_j = 1, 2, \dots, k_j - 2.$$

Waiting states

$$\pi_{t+1}(x + e_j, j, -2) = \pi_t(x, j, k_j - 1) \cdot p_j(t + 1).$$

Achievement of developing a run

$$\pi_{t+1}(x, j', 1) = \pi_t(x, j, -2) \cdot p_{j'}(t + 1), \quad j' \neq j,$$

$$\pi_{t+1}(x, 0, 0) = \pi_t(x, j, -2) \cdot q(t + 1).$$

Overflow states

$$\pi_{t+1}(x - e_j, j, -1) = \pi_t(x, j, -2) \cdot p_j(t + 1).$$

Continuing overflow states

$$\pi_{t+1}(x, j, -1) = \pi_t(x, j, -1) \cdot p_j(t + 1).$$

Termination of developing a run

$$\pi_{t+1}(x, j, 1) = \sum_{j \neq j'} \sum_{i_j=-2}^{k_j-1} \pi_t(x, j, i_j) \cdot p_{j'}(t + 1),$$

$$\pi_{t+1}(x, 0, 0) = \sum_{j=1}^c \sum_{i_j=-2}^{k_j-1} \pi_t(x, j, i_j) \cdot q(t + 1).$$

The distribution of E can be given with the state distribution of Y_n as

$$\Pr\{E = x\} = \sum_{(x, (j, i_j)) \in C_x} \pi_n(x, j, i_j).$$

3. Numerical examples

3.1. Fundamental conditions

In the section 2, the transition process of the state distribution of Y_t in $(c+1)$ -state trials has been shown. In this section we give a numerical examples of the joint distribution for a random vector $\mathbf{G} = (G_{n,k_1}^{(1)}, G_{n,k_2}^{(2)})$ as mentioned in Section 1. For simplicity, we discuss the joint distribution of two kinds of success runs in three-state trials. Doi et al. (1996) showed other types of examples of the joint distribution of the success and failure runs for N and $\mathbf{G}$ in two-state and three-state trials. To show transition processes of the state distribution of Y_t , we will construct a probability table. The dimension of the table can be determined by the number of cells in the partition $\{C_x\}$ and elements of $\mathbf{J}$. Therefore, the dimension of the table is $\#(\{C_x\}) \times \#(\mathbf{J})$ for any statistic vector $\mathbf{X} = (X_{n,k_1}^{(1)}, X_{n,k_2}^{(2)}, \dots, X_{n,k_c}^{(c)})$.

We discuss transition processes of the joint distribution for two kinds of success runs in a trinary sequence of length 3. It is assumed that each of the runs are accepted when $k_1 = 2$ and $k_2 = 3$ and each of the trials are time-homogenous. Summarizing above, $\mathbf{G} = (G_{3,2}^{(1)}, G_{3,3}^{(2)})$, $P(S_1|t) = p_1(t) = p_1$, $P(S_2|t) = p_2(t) = p_2$ and $P(F|t) = q(t) = q (= 1 - (p_1 + p_2))$.

3.2. The example of $\mathbf{G}$

In the case of $\mathbf{G} = (G_{3,2}^{(1)}, G_{3,3}^{(2)})$, the state space is defined by $\Omega = \{C_{(0,0)}, C_{(1,0)}, C_{(0,1)}, C_{(1,1)}\}$, where $\Xi = \{(0,0), (1,0), (0,1), (1,1)\}$, $C_{(x_1, x_2)} = \{((x_1, x_2), (j, i_j)) : (j, i_j) \in \mathbf{J}\}$, $\mathbf{J} = \{(0,0), (1,1), (2,1), (2,2), (1,-1), (2,-1)\}$ and states $((0,0), (1,-1)), ((0,0), (2,-1)), ((0,1), (1,-1)), ((1,0), (2,-1))$ are excluded from Ω . Since $\#(\{C_x\}) = 4$ and $\#(\mathbf{J}) = 6$ we can construct a (4×6) -dimensional probability table to calculate the joint distribution of $\mathbf{G} = (G_{3,2}^{(1)}, G_{3,3}^{(2)})$.

Table 1 at the initial time ($t = 0$) is defined as follows:

The transition processes of probabilities are shown in tables 2–5:

The first trial ($t = 1$):

The second trial ($t = 2$):

The last trial ($t = 3$):

Subsequently, the objective probability $\Pr\{\mathbf{G} = \mathbf{x}\}$ is shown below:

$$\Pr\{\mathbf{G} = (0,0)\} = p_1^2 p_2 + p_2^2 q + 3 p_1 p_2^2 + 3 p_2^2 q + 6 p_1 p_2 q + 3 p_1 q^2 + 3 p_2 q^2 + q^3,$$

Table 1

	$C_{(0,0)}$	$C_{(1,0)}$	$C_{(0,1)}$	$C_{(1,1)}$
(0,0)	1	0	0	0
(1,1)	0	0	0	0
(2,1)	0	0	0	0
(2,2)	0	0	0	0
(1,-1)	—	0	—	0
(2,-1)	—	—	0	0

Table 2

	$C_{(0,0)}$	$C_{(1,0)}$	$C_{(0,1)}$	$C_{(1,1)}$
(0,0)	q	0	0	0
(1,1)	p_1	0	0	0
(2,1)	p_2	0	0	0
(2,2)	0	0	0	0
(1,-1)	—	0	—	0
(2,-1)	—	—	0	0

Table 3

	$C_{(0,0)}$	$C_{(1,0)}$	$C_{(0,1)}$	$C_{(1,1)}$
(0,0)	$p_1 q + p_2 q + qq$	0	0	0
(1,1)	$p_2 p_1 + qp_1$	0	0	0
(2,1)	$p_1 p_2 + qp_2$	0	0	0
(2,2)	$p_2 p_2$	0	0	0
(1,-1)	—	0	—	0
(2,-1)	—	—	0	0

Table 4

	$C_{(0,0)}$
(0,0)	$p_1 p_2 q + p_1 qq + p_2 p_1 q + p_2 p_2 q + p_2 qq + qp_1 q + qp_2 q + qq q$
(1,1)	$p_1 p_2 p_1 + p_1 qp_1 + p_2 p_2 p_1 + p_2 qp_1 + qp_2 p_1 + qpq p_1$
(2,1)	$p_1 qp_2 + p_2 p_1 p_2 + p_2 qp_2 + qp_1 p_2 + qpq p_2$
(2,2)	$p_1 p_2 p_2 + qp_2 p_2$
(1,-1)	—
(2,-1)	—
$\Pr\{G = x\}$	$\Pr\{G = (0,0)\}$

Table 5

	$C_{(1,0)}$	$C_{(0,1)}$	$C_{(1,1)}$
(0,0)	$p_1 p_1 q$	0	0
(1,1)	0	0	0
(2,1)	$p_1 p_1 q$	0	0
(2,2)	0	0	0
(1,-1)	$p_1 p_1 p_1 + p_2 p_1 p_1 + qp_1 p_1$	—	0
(2,-1)	—	$p_2 p_2 p_2$	0
$\Pr\{G = x\}$	$\Pr\{G = (1,0)\}$	$\Pr\{G = (0,1)\}$	$\Pr\{G = (1,1)\}$

$$\Pr\{G = (1,0)\} = p_1^3 + 2p_1^2 p_2 + 2p_1^2 q,$$

$$\Pr\{G = (0,1)\} = p_2^3,$$

$$\Pr\{G = (1,1)\} = 0.$$

Acknowledgements

We wish to thank the editor and the referee for careful reading of our manuscript and helpful suggestions which led to improved results.

References

- Doi, M., Kanaya, T., Yamamoto, E., 1996. Joint distribution of success and failure runs. Tech. Report of Okayama Statistical Association, No. 61.
- Feller, W., 1968. An Introduction to Probability Theory and Its Applications, vol. I, 3rd ed. Wiley, New York.
- Fu, J.C., 1996. Distribution theory of runs and patterns associated with a sequence of multi-state trials. *Statist. sinica* 6, 957–974.
- Fu, J.C., Koutras, M.V., 1994. Distribution theory of runs: a Markov chain approach. *J. Am. Statist. Assoc.* 89, 1050–1058.
- Ling, K.D., 1988. On binomial distributions of order k . *Statist. Probab. Lett.* 6, 247–250.
- Mood, A.M., 1940. The Distribution Theory of Runs. *Ann. Math. Statist.*, 11, 367–392.
- Tsutsui, K., Tsutsui, K., Okada, S., Watarai, S., Seki, S., Yasuda, T., Shohmori, T., 1993. Identification and characterization of a nuclear scaffold protein that binds the matrix attachment region DNA. *The J. Bio. Chem.* 268 (17) 12886–12894.
- Wald, A., Wolfowitz, J., 1940. On a test whether two samples are from the same population. *Ann. Math. Statist.* 11, 147–162.